Project 02 — DevOps

Team Task Division

YAS: Yet Another Shop — CI/CD & Service Mesh Deployment

Deadline: 22/04/2026 | Group of 4 Members

✅ What Already Exists in the Repo (Reuse These!)

The nashtech-garage/yas repository provides significant infrastructure out of the box. The team should leverage these assets and NOT rewrite them from scratch.

Asset	Details	Status
Dockerfiles	All 20 services have Dockerfiles ready (cart, tax, product, order, payment, customer, inventory, location, media, rating, recommendation, promotion, search, webhook, sampledata, storefront, backoffice, backoffice-bff, storefront-bff, payment-paypal)	✅ **Ready to use**
GitHub Actions CI	22 existing workflows — one per service. Already builds & pushes images to ghcr.io on main branch push. Files in .github/workflows/	✅ **Extend only**
Helm Charts	Full Helm charts for all 24 services in k8s/charts/. Values.yaml pre-configured with correct image names.	✅ **Ready to use**
Cluster Setup Scripts	k8s/deploy/ contains: setup-cluster.sh, setup-keycloak.sh, setup-redis.sh, deploy-yas-applications.sh — sets up Postgres, Kafka, Elasticsearch, Keycloak, Redis, Observability stack automatically.	✅ **Run directly**
Minikube Guide	k8s/deploy/README.md — complete step-by-step guide including Minikube setup, /etc/hosts entries, and all domain mappings.	✅ **Follow guide**

❌ What the Team Must Build from Scratch

Missing Piece	Requirement	Assigned To
Branch CI: commit-ID image tag + Docker Hub push	Req. 3 — On any branch push, build image tagged with latest commit SHA and push to Docker Hub	**Member 2**
Jenkins: developer_build job	Req. 4 — Parameterized job per service branch, deploys correct image to K8s, exposes NodePort URL	**Member 3**
Jenkins: teardown job	Req. 5 — Cleans up developer_build deployment via helm uninstall	**Member 3**
ArgoCD dev/staging (Advanced)	Advanced +2pts — Auto-sync dev namespace on main; staging on release tag v*.*.*	**Member 4**
Istio mTLS + Kiali (Advanced)	Advanced +2pts — Enable mTLS, AuthorizationPolicy, retry VirtualService, Kiali topology	**Member 4**

👥 Team Task Division (4 Members)

Member 1 — Infra & K8s Lead

Cluster setup + Helm deployment

- Install Minikube (requirement: 16GB RAM, 40GB disk, Ubuntu/WSL2)
- Run setup-keycloak.sh -> setup-redis.sh -> setup-cluster.sh in order
- Configure cluster-config.yaml with team domain, passwords, and replica counts
- Run deploy-yas-configurations.sh then deploy-yas-applications.sh to deploy all services
- Add NodePort services for developer access (storefront, backoffice, swagger-ui)
- Configure /etc/hosts with correct IP mappings (run: minikube ip)
- Create dev and staging namespaces for pipeline targets
- Verify full app is accessible end-to-end before team proceeds

Member 2 — CI Pipeline Lead

GitHub Actions + Docker Hub integration

- Fork nashtech-garage/yas to the team GitHub account
- Add Docker Hub credentials as GitHub Secrets: DOCKERHUB_USERNAME, DOCKERHUB_TOKEN
- Modify all 22 CI workflow files to trigger on push to any branch (not just main)
- Add step to each workflow: build image tagged with github.sha, push to Docker Hub
- Keep existing latest tag push on main branch — do not remove it
- Create a shared composite action (.github/workflows/actions/) for Docker Hub login to avoid repeating code
- Test: create a feature branch, commit a change, verify image appears on Docker Hub with correct SHA tag

- Document the new Docker Hub image naming convention for the team

Member 3 — Jenkins CD Lead

Jenkins jobs + developer_build pipeline

- Deploy Jenkins on K8s cluster (Helm chart) or as Docker container on worker node
- Install required Jenkins plugins: Kubernetes, Pipeline, Git, Docker, Blue Ocean
- Create developer_build pipeline job with per-service branch parameters
- Job logic: look up commit SHA for the specified branch -> use that SHA image tag for target service, latest for all others
- Display a clickable NodePort URL in the Jenkins build result page
- Create teardown pipeline job that runs helm uninstall for all services in the yas namespace
- Set up GitHub webhook to auto-trigger Jenkins on main branch changes
- Write Jenkinsfile for both jobs and commit to team repo

Member 4 — Advanced Features Lead

ArgoCD + Istio Service Mesh

- [ArgoCD] Install ArgoCD on K8s cluster using official Helm chart
- [ArgoCD] Create Application manifest for dev namespace — auto-sync on main branch changes
- [ArgoCD] Create Application manifest for staging — sync only on release tag matching v*.*.*
- [Istio] Install Istio (istioctl) + Kiali dashboard on the K8s cluster
- [Istio] Enable mTLS for all YAS services using PeerAuthentication and DestinationRule resources
- [Istio] Create AuthorizationPolicy — allow only permitted service-to-service communication pairs
- [Istio] Create VirtualService with retry policy: on HTTP 500 response, retry up to 3 times
- [Testing] kubectl exec into test pods, curl between services, verify allow/deny behavior
- [Deliverables] Screenshot Kiali service topology, collect all YAML manifests, write README

🤝 Shared Responsibilities (All 4 Members)

- Write the .docx report with screenshots of every configuration step
- Integration testing: verify developer_build job works end-to-end with a real service branch
- Peer review each other's work before submission
- Prepare demo walkthrough for grading session
- Submit by 22/04/2026 — filename: MSSV1_MSSV2_MSSV3_MSSV4.docx (sorted ascending)

📋 Quick Reference Summary

Member	Role	Key Deliverable	Requirements
Member 1	Infra & K8s Lead	Minikube cluster fully running with all YAS services deployed via Helm	Req. 1, 2
Member 2	CI Pipeline Lead	22 workflows extended to push commit-SHA images to Docker Hub on any branch	Req. 3

Member	Role	Key Deliverable	Requirements
Member 3	Jenkins CD Lead	developer_build job + teardown job working in Jenkins with NodePort output	Req. 4, 5
Member 4	Advanced Lead	ArgoCD dev/staging + Istio mTLS + Kiali topology + test logs	**Advanced +4 pts**

GitHub: https://github.com/nashtech-garage/yas